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Simple USB Interface SIGNAL MONITOR

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Presented here is a simple, low-cost USB interface signal monitoring circuit. It can be used to monitor signal activity for most of the USB interfaces.

The circuit finds following applications:

1. It can be connected to a cable with USB interface to check data signals in the wires
2. It can be connected between two USB cables, with the first cable connected to a computer and the second to a peripheral device with a USB interface, to monitor various signals
3. It can be connected to a USB power supply or USB battery charger and other devices with USB interfaces to check power signals
4. It can also be connected to a voltmeter or an oscilloscope to check voltage signals

In all these cases, the maximal USB cable length should be limited as specified under the USB standard.

Circuit and working

Circuit diagram of the USB interface signal monitor is shown in Fig. 1. It is built around two BC557 pnp transistors (T1 and T2), three BC547 npn transistors (T3 through T5), seven LEDs (LED1 through LED7), two USB connectors (CON1 and CON2), and a few resistors and capacitors.

Connect one of the two USB connectors to a computer, and the other to a peripheral with USB interface. LED6 and LED7 show the data bus status. You can also use voltmeters, oscilloscopes and other test equipment to monitor the USB interface activity.

LED Operation table details the LEDs' indication in the circuit. Voltage drop across all the LEDs should be around 1.6V, and the LEDs should be able to produce light at current levels as low as 1mA. Resistors R1, R2 and R9 through R11 are used to limit the

LED OPERATION

LED	Interface status
LED1 = On	Data- = Low level
LED2 = On	Data+ = Low level
LED3 = On	Data- = High level
LED4 = On	Data+ = High level
LED5 = On	Data+ and Data- are high at the same time
LED6 = On	The LED is 'on' if you have normal power supply
LED7 = On	The LED is 'on' if you have reverse power supply

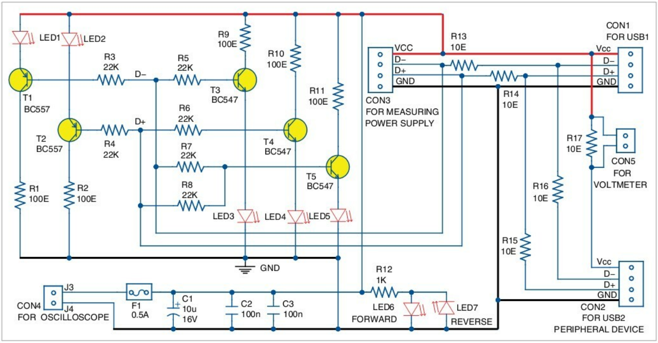

Fig. 1: Circuit diagram of the simple USB interface signal monitor